RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 15 October 2026 – Telemetry Fault Recovery

2. DAILY LOG ENTRY FORM

- **Date:** 15 October 2026

• 1. Assigned Task:

Telemetry Fault Recovery — corrupted and missing telemetry handling and interpolation logic**.**

• 2. What Was Studied:

Studied missing and corrupted UAV telemetry readings and their effects on data processing.

Reviewed sensor range validation for temperature, pressure, altitude, and speed.

Also studied how linear interpolation can estimate suitable missing values using surrounding valid readings.

• 3. What Was Implemented:

Created and ran a Python telemetry fault recovery routine using Pandas and NumPy.

Loaded the Day-7 synthetic UAV telemetry dataset when available and introduced controlled test faults.Implemented range validation, invalid-value handling, and linear interpolation for internal gaps.

Exported the recovered telemetry and a JSON summary containing recovery statistics.

- **4. Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.14.6, Pandas, NumPy, pathlib, JSON, CSV.

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- **5. Output File Name-** suraj_rudra_task/ Day-11/codes/telemetry_fault_recovery.py
  Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**
  No problem was encountered

- **7. How Problem Was Resolved:**
  No problem was encountered.

- **8. Next Task Scheduled:**

  (15 October 2026)- Final Documentation & package –Final documentation and software/data readiness package.

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 15/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026